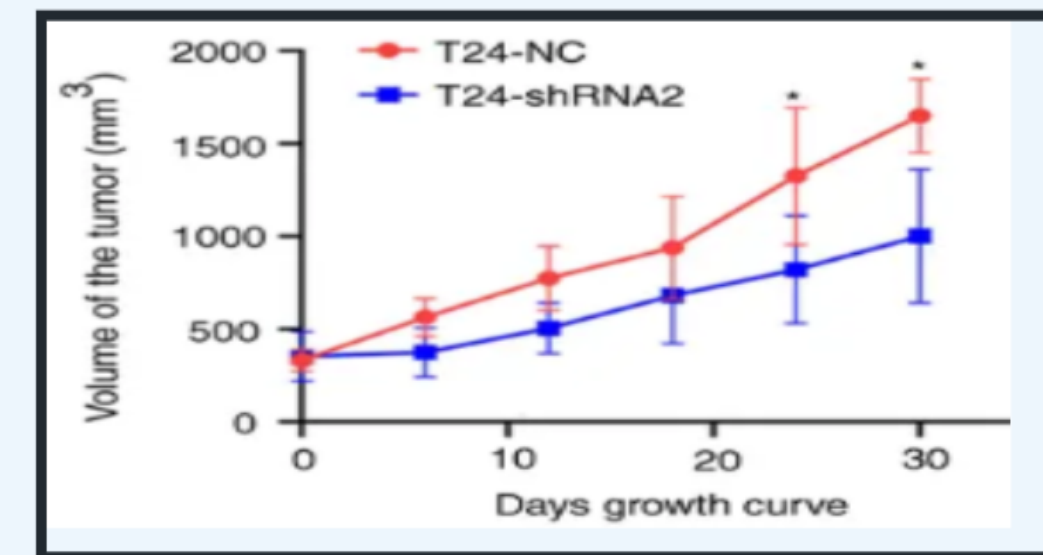


Task Taxonomy

Panel-Level Fine-Grained Perception

Numerical Perception



Difference of red and blue curve
 ≈ 700 in day30

Slope of increase red curve
 ≈ 45

Morphological Perception



Cell



Micro-organism



Sub-cellular



Tissue

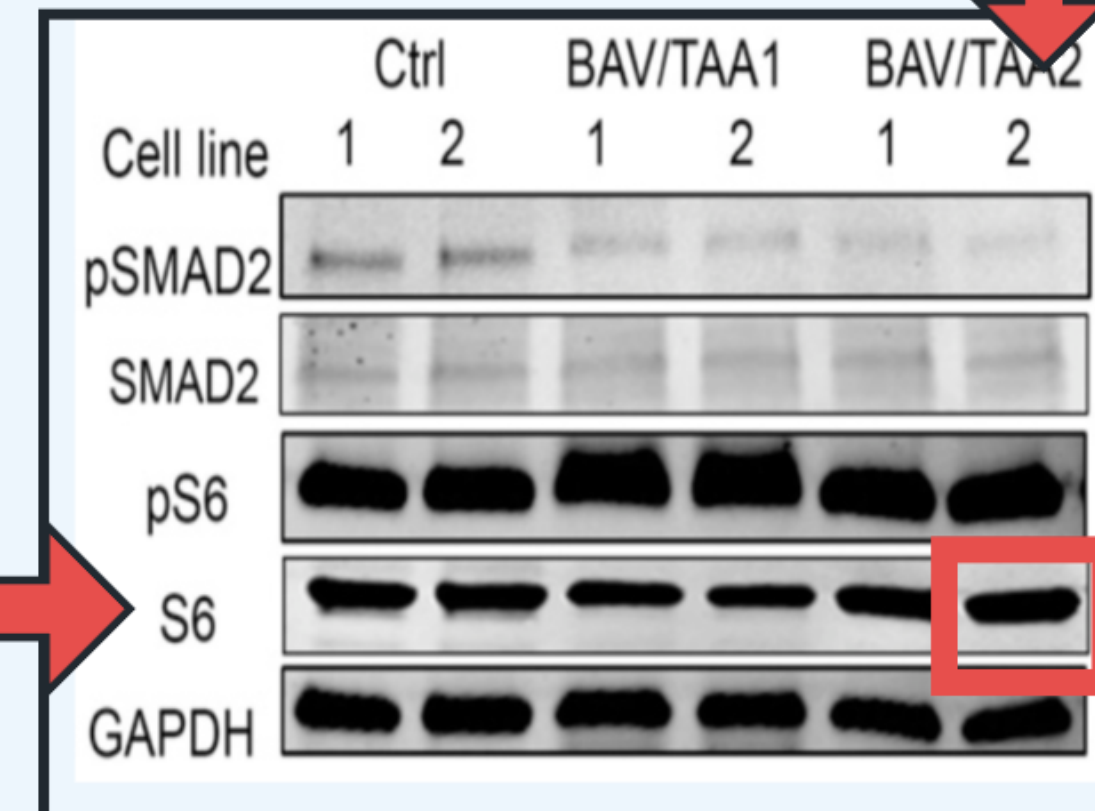
Elliptical Cell Nucleu

Cells exhibit tight clustering

Staining area 30%

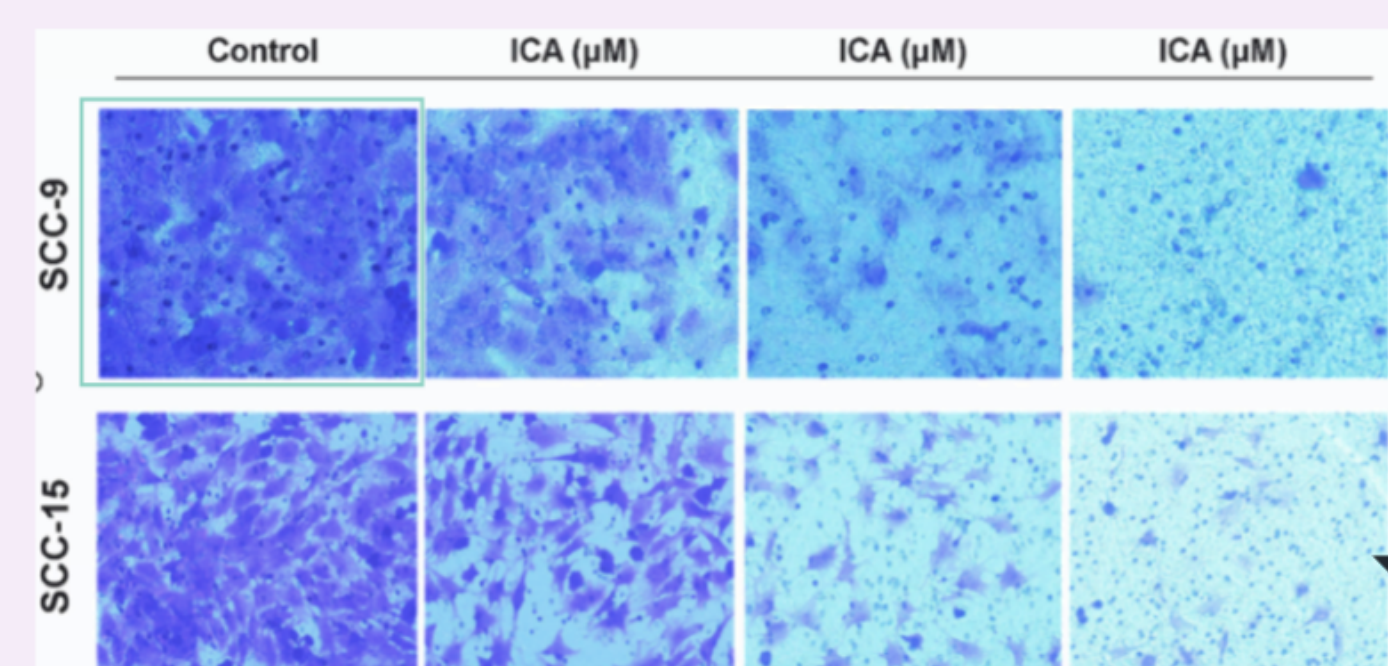
Information Localization

Experiment Condition
S6 + BAV/TAA2



Cross-Panel Relation Understanding

Trend Analysis



Trend is consistent in 2 groups

Parallel Comparative Experiment

Degree of infection decline

Heterogeneous Integration

Heterogenous: Western Blot & Chart

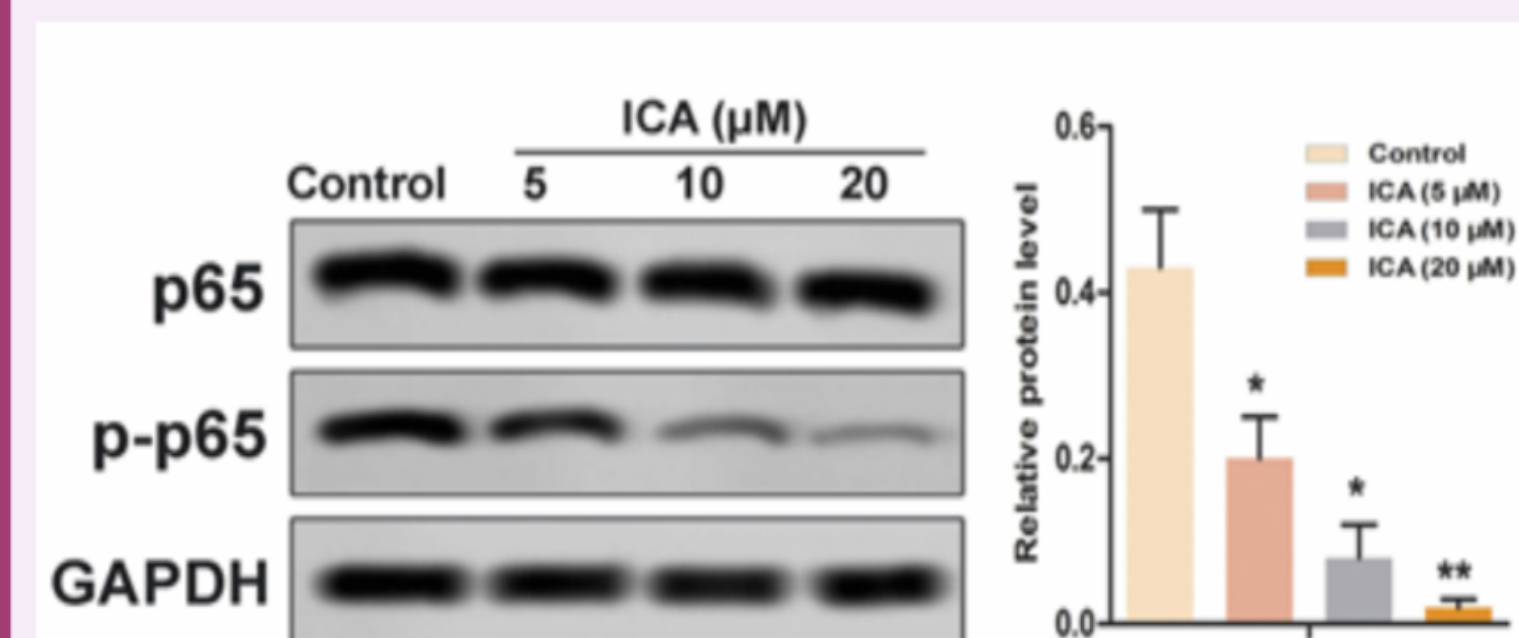


Chart is used to assist in displaying the values of WB

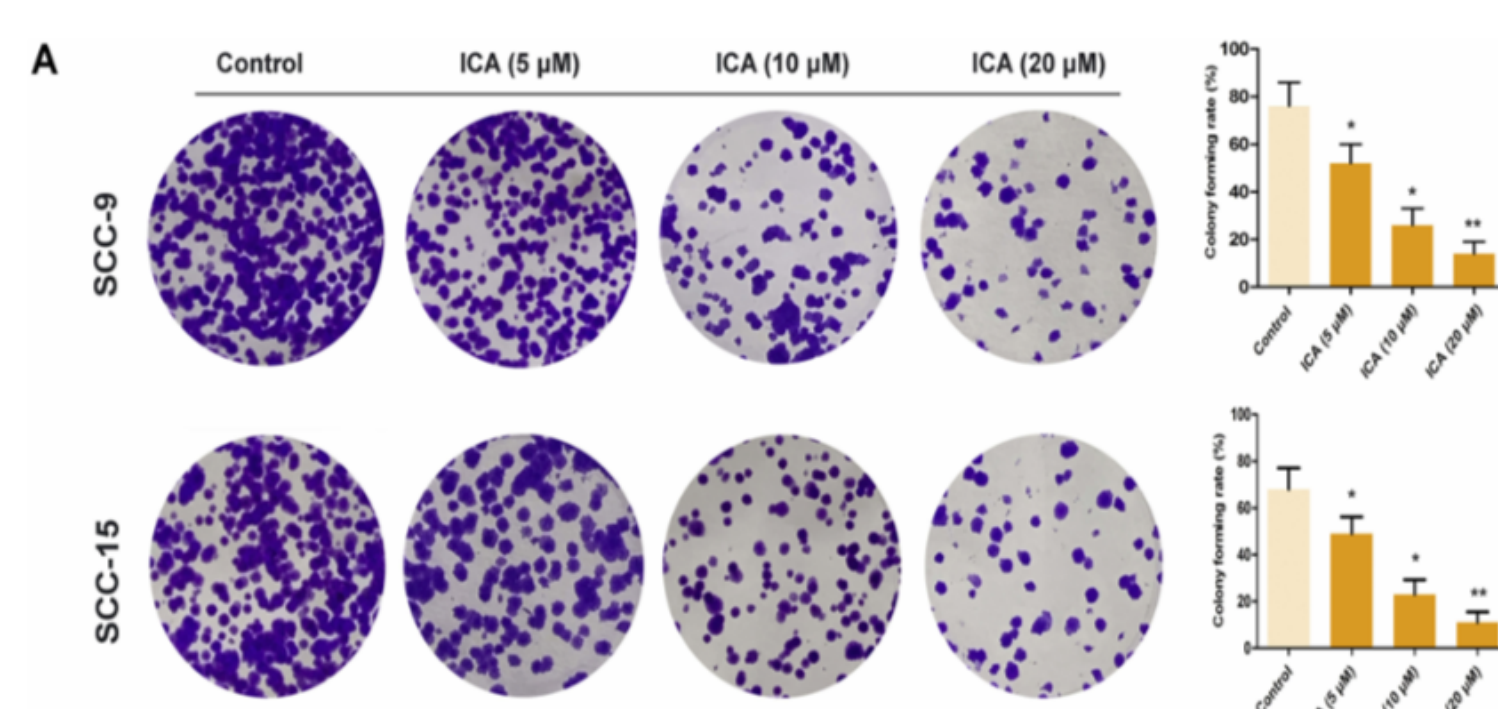
Expert-Level Reasoning

Qualitative Reasoning

Cell Experiments

Evaluate the regulatory effects

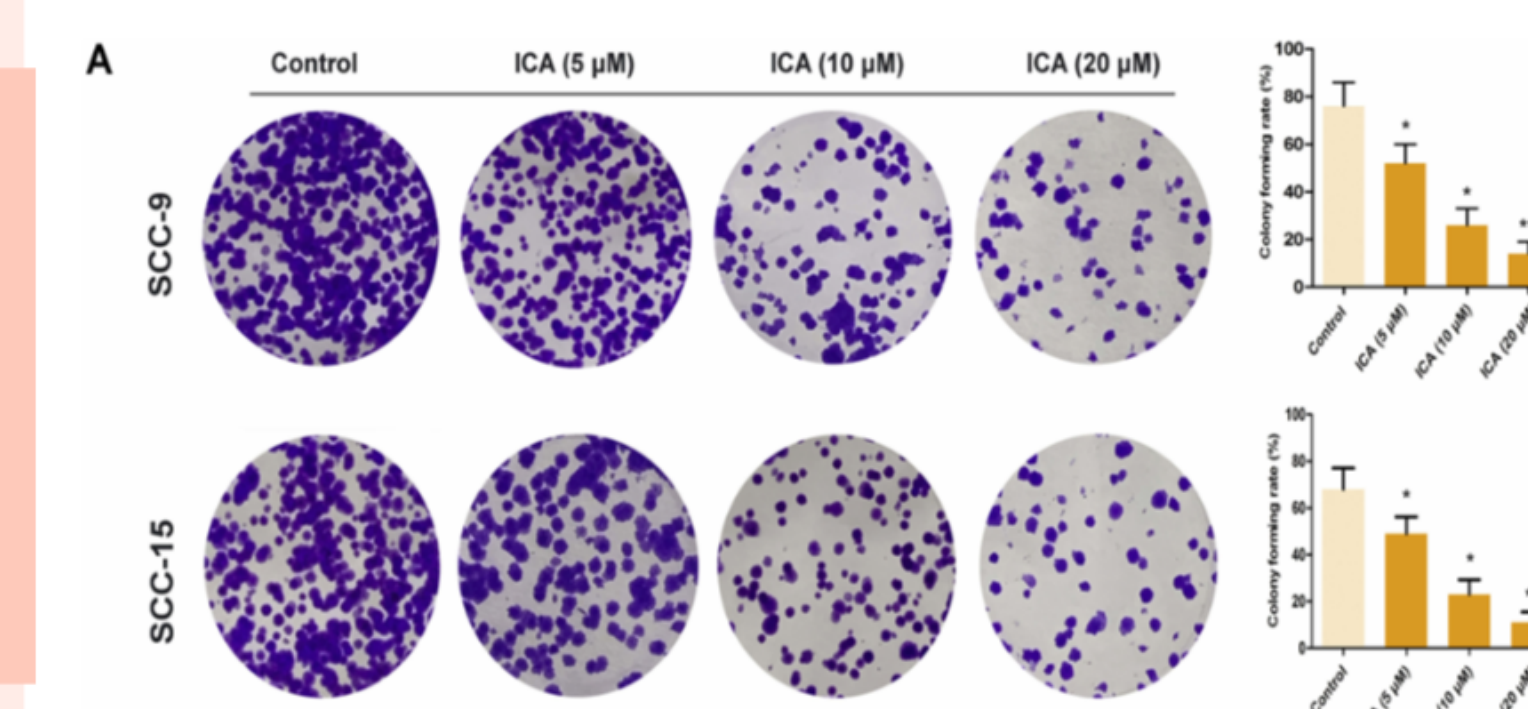
ICA inhibits the proliferative capacity



Quantitative Reasoning

Inhibition rate of the ICA (5 μM) group $\approx 30\%$

Inhibition rate of ICA (10 μM) group $\approx 60\%$

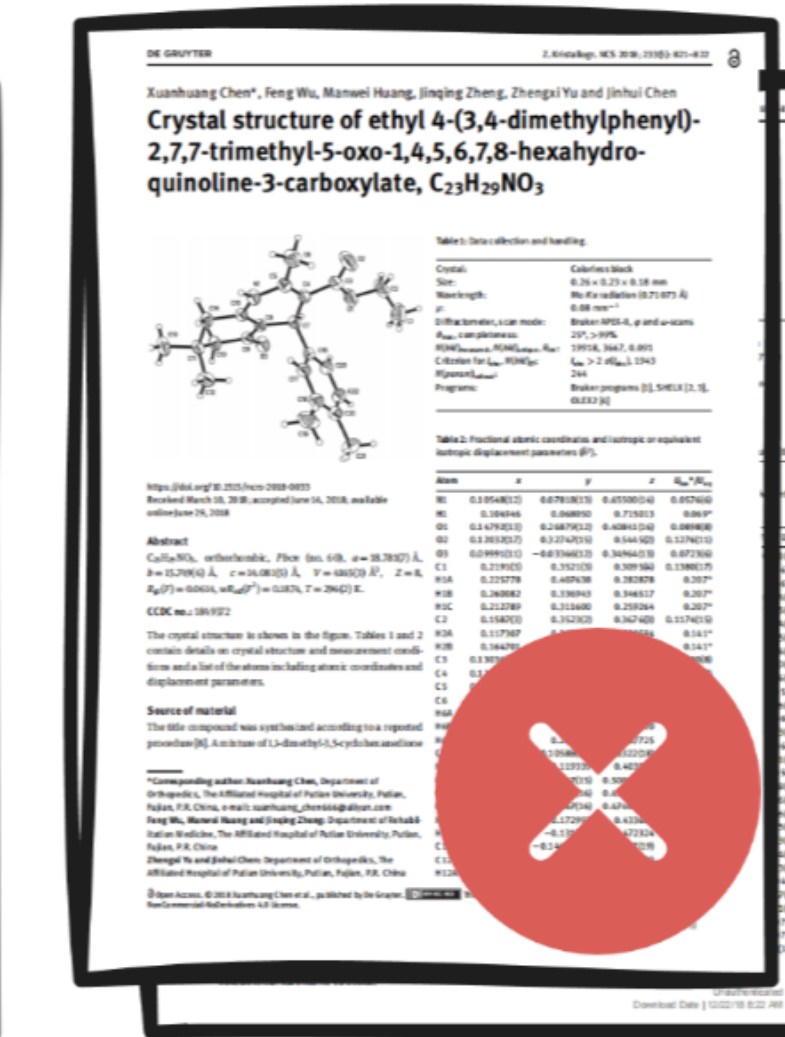
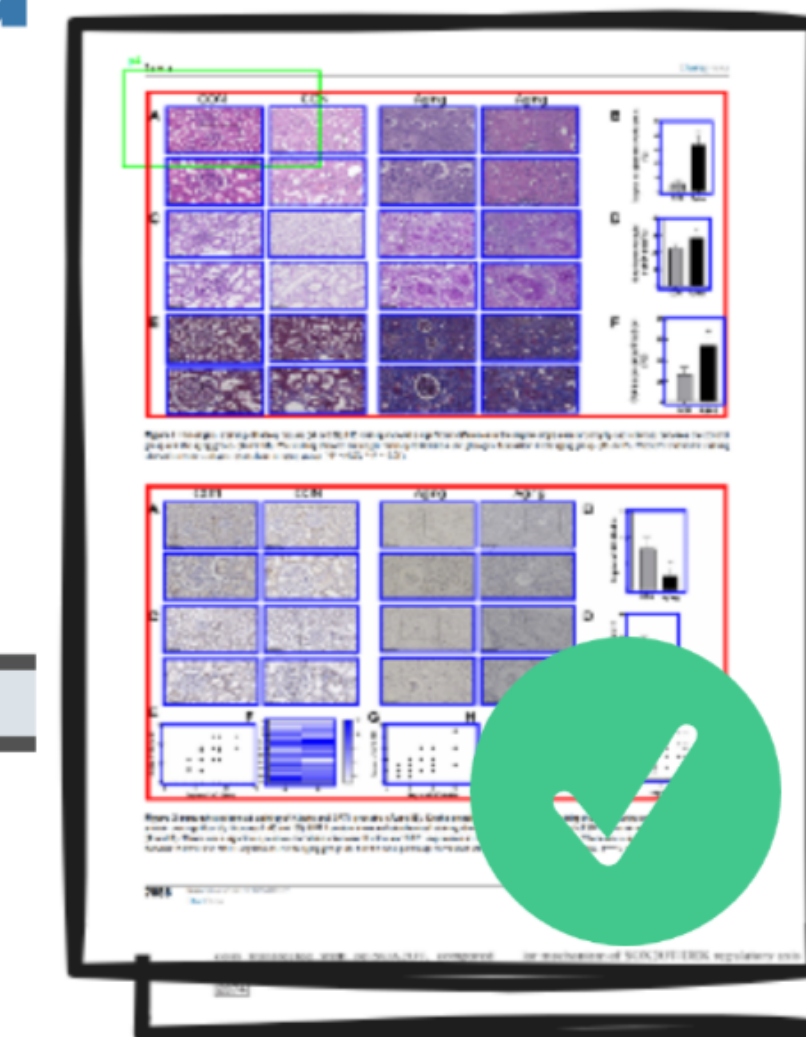


1. Image Collection

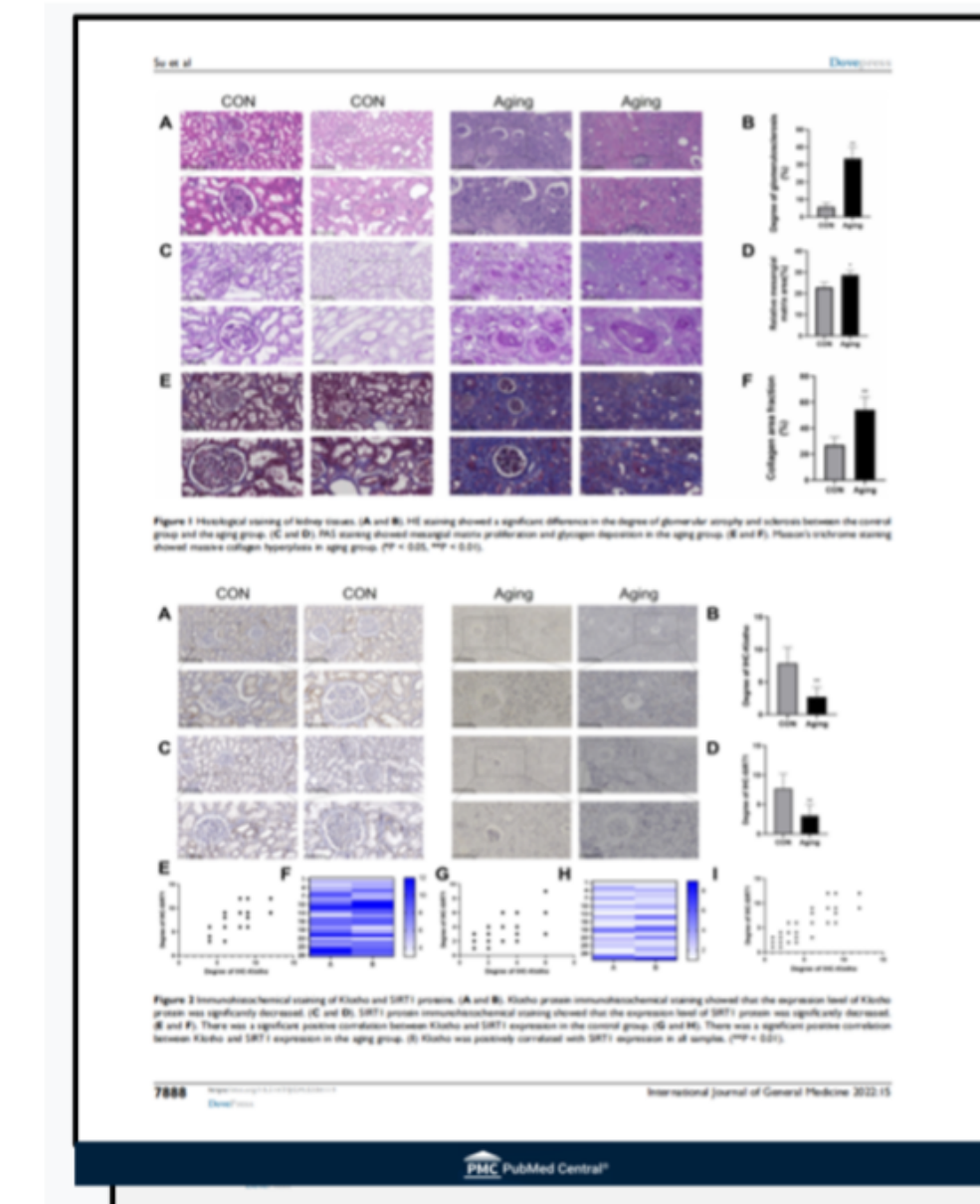
Papers

PubMed

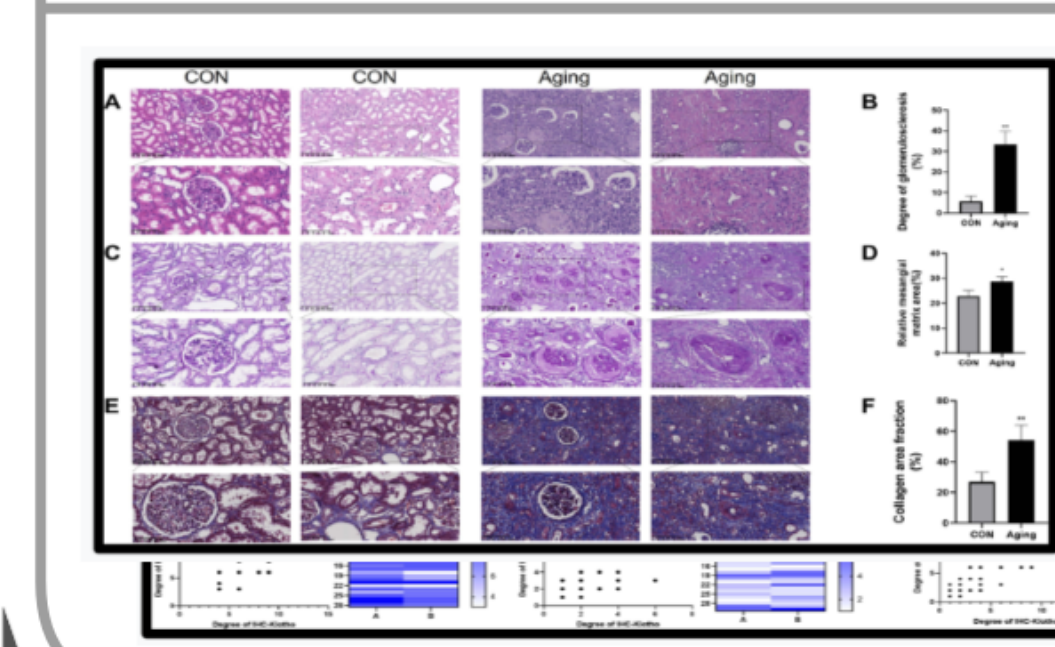
5000+



PDF Pages



Images



Captions

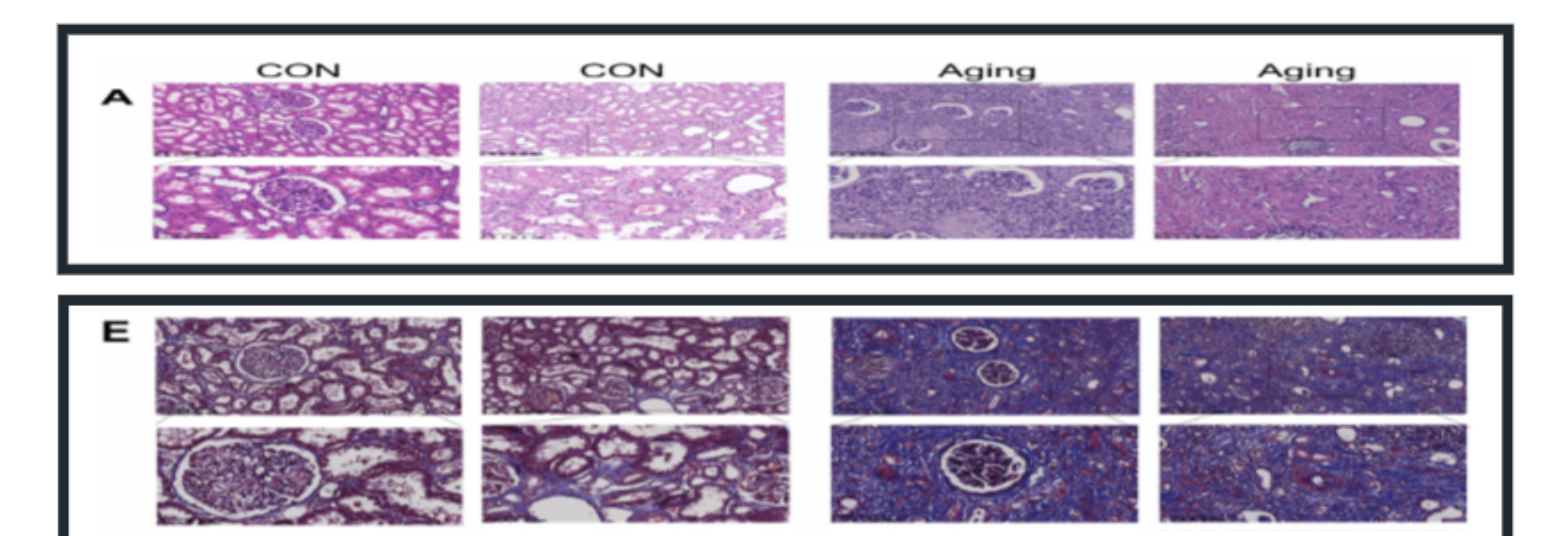
Immunohistochemical staining of Klotho and SIRT1 proteins. (A and B). Klotho protein immunohistochemical...

Related Sentences

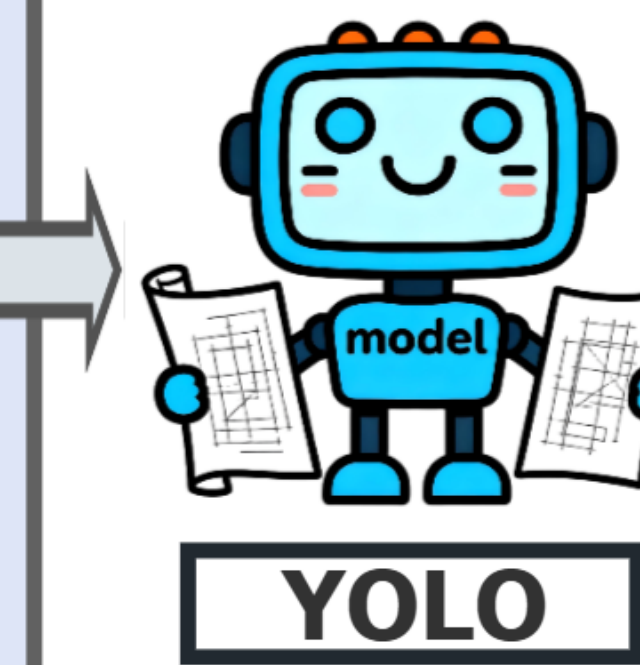
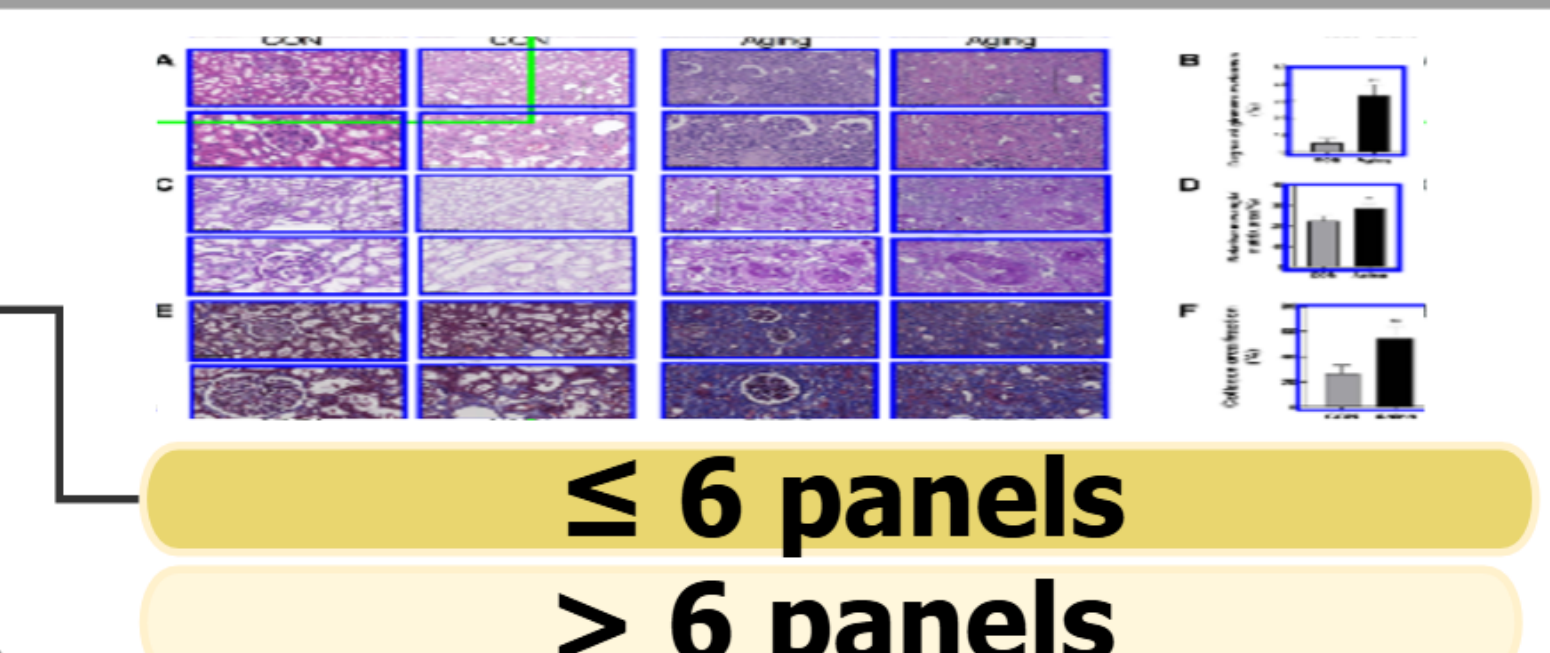
The localization and expression intensity of Klotho and SIRT1 in kidney tissue were detected by immunohistochemistry...

2. Image Filtering & Curation

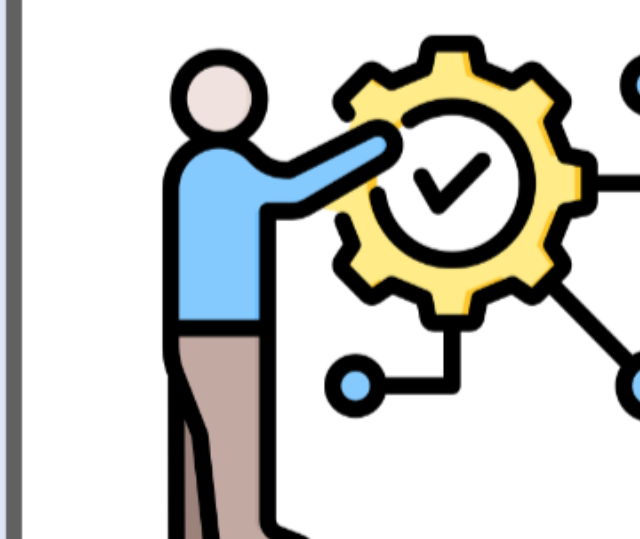
Panels



Figures



Complexity



Expert Correction



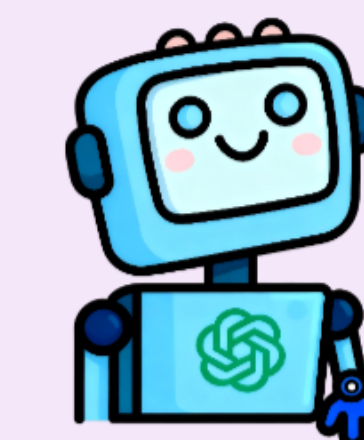
Quality

Completeness

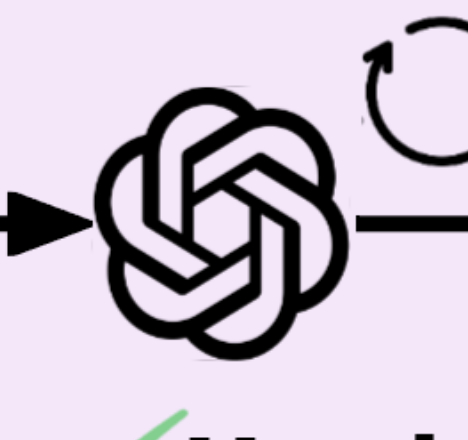
Integrity

Representativeness

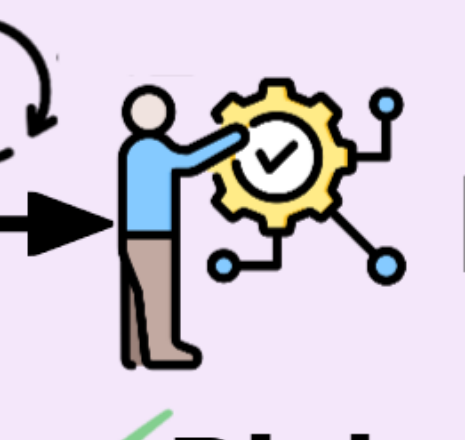
3. Question Generation



QA



✓ Hard
✗ Easy



✓ Right
✗ Error



Question

In the experiment, different samples (FFPE, capture frozen, polyA frozen, etc.) were tested. Combined with the information in Figures A-D, which of the following inferences about sample characteristics and data correlation is correct?

- A. In Figure A, the higher the proportion of rRNA fragments in polyA frozen samples, the higher aligned fragments in Figure B.
- B. For Figure E, the more severe the RNA degradation, the higher the proportion of duplicate fragments in capture libraries.
- C. It can be inferred that sample processing methods affect the statistical results of the proportion of rRNA fragments.
- D. If a sample has a large fragment diversity value in Figure C, then the rRNA of that sample in Figure A must be at a low level.